DIGITAL FORENSICS LAB: INTERACTIVE CYBERSECURITY INCIDENT INVESTIGATION AND EVIDENCE ANALYSIS SIMULATOR

A MINI-PROJECT REPORT

Submitted in partial fulfilment of the award of the degree of

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IN

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Submitted by:

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BONAFIDE CERTIFICATE

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ABSTRACT

In modern enterprise computing environments, cyber attacks such as credential stuffing, unauthorized data exfiltration, system tampering, and lateral movement have surged dramatically. However, traditional computer science education and corporate security training often present digital forensics and incident response (DFIR) as static, abstract theoretical concepts or rely on prohibitive, heavy virtual machines that are complex to configure and lack intuitive educational scaffolding.

To bridge this crucial gap, this project presents **Digital Forensics Lab (DFIR Lab)**, a high-fidelity, interactive, browser-based cybersecurity incident investigation and evidence analysis simulator. Built with a dependency-free vanilla architecture (HTML5, modern CSS3 with custom properties, and modular ES6+ JavaScript), the platform allows aspiring security analysts, students, and practitioners to perform authentic forensic investigations directly within any modern web browser.

The platform simulates realistic incident dossiers across three complex attack scenarios: unauthorized credential misuse (*The Midnight Access*), data destruction and anti-forensics (*The Missing Files*), and executive account compromise via rogue hardware (*The Unknown Device*). Each case provides complete forensic artifacts, including multi-source raw logs (SSO authentication, syslog, bash command histories, reverse proxy HTTP headers, and file system metadata). Users can verify cryptographic chain-of-custody checksums using SHA-256 hashes, reconstruct event sequences in an interactive chronological timeline, flag suspicious indicators of compromise (IOCs), document working hypotheses in an integrated investigator notebook, and test their analytical reasoning in a scenario-driven examination. Upon completion, the system automatically synthesizes verified technical facts into a standardized, audit-ready Incident Investigation Report.

By eliminating complex software dependencies and delivering real-time sensory feedback, intuitive plain-English log translations, and professional forensic workflows, the Digital Forensics Lab dramatically lowers the barrier to learning cyber forensics while maintaining forensic rigor and scientific standards.

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CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

Digital forensics and incident response (DFIR) has emerged as one of the most critical pillars of modern information security. Whenever an enterprise organization experiences a data breach, malware intrusion, unauthorized access, or internal data sabotage, forensic investigators are summoned to reconstruct the timeline of compromise, identify the threat actor's tactics, techniques, and procedures (TTPs), preserve digital evidence under strict chain of custody, and remediate vulnerabilities.

Despite the critical need for qualified digital forensic practitioners, learning digital forensics remains fraught with hurdles. Aspiring students and junior analysts are typically subjected either to dry theoretical textbooks devoid of hands-on log triage, or to heavyweight, complex capture-the-flag (CTF) environments that require gigabytes of disk image downloads, intricate hypervisor installations, and advanced proprietary software suites (such as EnCase, FTK, or commercial SIEM instances).

The **Digital Forensics Lab** addresses these challenges by delivering an interactive, lightweight, zero-dependency, web-based digital forensics workstation simulator. Through a single, modern web interface, users step into the role of a certified forensic investigator, examining realistic evidence packets, correlating authentication events against file system changes, uncovering anti-forensics measures, proving digital integrity using cryptographic hashes, and generating formal incident findings.

1.2 SCOPE OF THE WORK

The scope of this project encompasses the design, architectural modeling, front-end implementation, forensic scenario authoring, and assessment validation of a complete digital investigation educational platform:

1. **Scenario-Based Simulation**: Authoring three distinct incident dossiers covering unauthorized external logins, internal evidence destruction, and unauthorized endpoint hijacking.

2. **Chain of Custody & Evidence Verification**: Implementation of SHA-256 cryptographic hashing to demonstrate proof of non-tampering in digital artifacts

according to ISO/IEC 27037 standards.

3. **Multi-Source Telemetry Correlation**: Parsing and displaying disparate log sources, such as SSO gateway logs, Apache/Nginx reverse proxy headers, Linux bash histories, syslog event logs, and file inode metadata.

4. **Interactive Timeline Modeling**: Dynamic chronological sequencing of events with filtering by threat severity (`critical`, `suspicious`, `informational`), log source, and keyword matching.

5. **Cognitive Scaffolding & Dual-Layer Representation**: Presenting raw technical log payloads alongside humanized plain-English explanations, guiding beginner learners toward professional log analysis competence.

6. **Assessment & Incident Reporting**: Providing scenario-specific objective validation questions, investigator rank progression, and an automated report generation engine that formats formal forensic reports.

1.3 PROBLEM STATEMENT

Traditional methods of educating computer science students in digital forensic analysis suffer from several critical shortcomings:

- **Steep Tooling Barrier**: Standard forensic labs require students to manage virtual machines, install Linux distributions, mount raw image files, and configure memory analyzers before analyzing a single log entry.
- **Lack of Narrative Context**: Many exercises present isolated command outputs without the narrative backdrop of a real corporate incident, divorcing evidence from realistic business impact.
- **Neglect of Chain of Custody & Legal Rigor**: Students frequently overlook the fundamental rule of forensics—distinguishing observed technical facts from premature assumptions of guilt, and documenting cryptographic integrity.
- **Absence of Real-Time Mentorship**: When beginners face cryptic syslog errors, hexadecimal dumps, or epoch timestamps, they lack inline contextual assistance, resulting in frustration and dropouts.

The Digital Forensics Lab resolves these issues by packaging complex multi-source telemetry, cryptographic integrity checks, interactive timeline manipulation, and

structured mentoring into an accessible, responsive web application.

1.4 AIM AND OBJECTIVES OF THE PROJECT

The overarching aim of this project is to create an engaging, technically rigorous, client-side digital forensics training platform that equips students with real-world incident investigation skills.

The specific objectives are:

1. To engineer a client-side Single-Page Application (SPA) using HTML5, modern CSS3, and ES6+ JavaScript requiring zero back-end server setup or database management.

2. To design an authentic cyber-themed user interface featuring dark mode aesthetics, responsive modular views, and dynamic visual indicators.

3. To develop three comprehensive forensic cases with curated JSON evidence datasets mirroring enterprise enterprise telemetry.

4. To integrate real-time cryptographic hash display and verification to reinforce the importance of evidence integrity in digital legal proceedings.

5. To build interactive investigation tools including a Chronological Event Timeline, Evidence Inspector, Red Flag Marker, and Investigator Notebook.

6. To implement an automated incident report generation module capable of compiling findings, conclusions, and remediations into an executive-ready document.

CHAPTER 2: SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

The Digital Forensics Lab is engineered to be lightweight, running entirely within modern web browser runtimes without requiring local virtualization, heavy memory allocations, or dedicated GPU accelerators.

Hardware Component	Minimum Requirement	Recommended Specification
Processor	Dual Core 1.8 GHz Intel / AMD / Apple Silicon	Quad Core 2.4 GHz or higher (Intel Core i5 / Apple M-Series)
System Memory (RAM)	2.0 GB RAM	4.0 GB RAM or higher
Storage (HDD / SSD)	50 MB free disk space	500 MB free SSD storage
Display Resolution	1024 x 768 (XGA)	1920 x 1080 (Full HD) or Retina
Input Devices	Standard Keyboard and Mouse / Trackpad	Standard Keyboard with Scroll Wheel Mouse

2.2 SOFTWARE SPECIFICATIONS

The system leverages standard web protocols and open web standards, ensuring platform independence across all modern operating systems.

Software Layer	Technology / Specification
Operating System	Platform Independent: Windows 10/11, macOS 12+, Ubuntu Linux 20.04+, ChromeOS
Target Runtime	Modern Chromium / WebKit / Gecko Browser (Google Chrome 90+, Mozilla Firefox 88+, Apple Safari 15+, Microsoft Edge 90+)

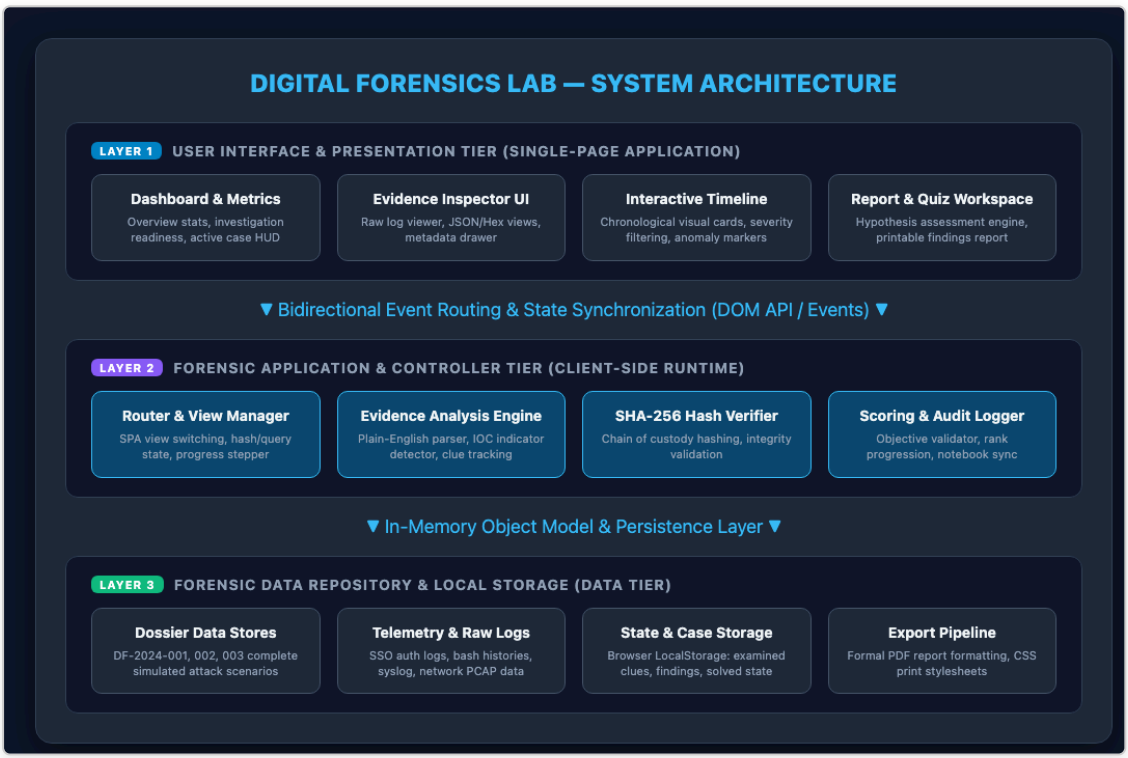
Software Layer	Technology / Specification
Front-End Presentation	Semantic HTML5, CSS3 Custom Properties (CSS Variables), Flexbox, CSS Grid, SVG Vector Icons
Application Logic	Pure Vanilla JavaScript (ECMAScript 2020+ / ES6+), Dependency-Free
Client-Side Storage	HTML5 Web Storage API (`window.localStorage` and `window.sessionStorage`)
Cryptographic Standards	W3C Web Cryptography API (`crypto.subtle` / SHA-256 Algorithm)
Export Format	Cascading Style Sheets Print Media Queries (`@media print`) and Standard PDF Export

---

CHAPTER 3: SYSTEM ARCHITECTURE

3.1 ARCHITECTURAL OVERVIEW

The architecture of the Digital Forensics Lab follows a clean, decoupled **Three-Tier Client-Side Architecture** designed for zero-latency interactivity, full data confidentiality (all data remains in the user's browser), and zero installation requirements.



System Architecture Diagram

Fig. 3.1: High-Level System Architecture of Digital Forensics Lab

3.2 TIER DECOMPOSITION

1. Presentation Tier (UI / UX Layer)

The presentation layer is responsible for rendering an immersive cyber-forensic workstation experience. It includes:

- **Forensic HUD & Dashboard**: Displays key metric cards (Active Cases, Evidence Examined, Solved Dossiers, Investigator Rank, Accuracy Index).
- **Navigation & View Router**: Implements smooth Single Page Application (SPA) tab switching between Dashboard, Case Files, Evidence Viewer, Timeline, Findings, Challenge, and Report.
- **Evidence Inspector Drawer**: Formats JSON telemetry and raw log payloads into syntax-highlighted, legible cards with plain-English toggle switches.
- **Event Timeline Matrix**: Dynamically constructs chronological log cards with severity badges, category pills, and quick-filter toolbars.
- **Investigator Notebook**: Provides a persistent modal notepad for recording hypotheses, timestamps, and suspicious IP addresses.

2. Application Logic Tier (Controller & Forensic Engine)

The middle tier encapsulates the forensic business logic and rules:

- **State Manager**: Tracks investigator progress across cases, records examined evidence IDs, and synchronizes state to `localStorage`.
- **Evidence Filtering & Search Engine**: Enables real-time text searching across log summaries, IP addresses, usernames, and file paths.
- **Cryptographic Hash Engine**: Validates SHA-256 signatures for every evidence item to guarantee chain of custody.
- **Assessment Evaluator**: Validates answers to scenario challenge questions, computes accuracy percentages, and determines rank promotions.
- **Report Synthesizer**: Aggregates verified evidence and findings into a printable, structured Incident Response Report.

3. Data Tier (Simulated Telemetry & Local Storage)

The data layer maintains the core forensic corpora:

- **Case Corpus (`FORENSIC_CASES`)**: Declarative in-memory structures storing scenario dossiers, difficulty levels, threat summaries, and mentor tips.
- **Evidence Log Store**: Multi-source telemetry representing authentic syslogs, auditd logs, bash histories, and network request records.

- **Browser Local Storage**: Persists user name, rank progress, exam completions, and custom investigator notes across page reloads.

CHAPTER 4: MODULE DESCRIPTION

4.1 CASE DOSSIER & SCENARIO SIMULATION ENGINE

This module is the entry point for forensic investigations. It maintains three fully fleshed-out incident scenarios:

1. **Case DF-2024-001: The Midnight Access** — An unauthorized late-night authentication and download of a confidential merger termsheet (`Project_Titan_Acquisition_Termsheet.pdf`) via compromised employee credentials.

2. **Case DF-2024-002: The Missing Files** — A data tampering and anti-forensics incident where finance spreadsheets were securely deleted and bash histories were scrubbed using elevated privileges.

3. **Case DF-2024-003: The Unknown Device** — An anomalous device accessing an executive account, bypassing standard MDM policies via remote session token hijacking.

The module manages case metadata, switches the active investigation context, and updates all child modules to reflect the active case's artifacts.

4.2 RAW EVIDENCE INSPECTOR & CRYPTOGRAPHIC VERIFICATION MODULE

Digital forensic integrity hinges upon the immutability of evidence. The Evidence Inspector provides:

- **Raw Telemetry Inspection**: Real-time rendering of authentic log formats (Linux syslog, Windows Security Event ID 4624/4625, Apache HTTP logs).
- **Cryptographic Hash Verification**: Every digital artifact is stamped with its SHA-256 hash. Analysts can inspect and verify these signatures to guarantee evidence has not been altered in transit.
- **Plain-English Explanations**: A pedagogical bridge translating complex log strings (e.g., `event_id:4625`, `pam_unix(sshd:auth): authentication failure`) into clear statements explaining what occurred.

4.3 MULTI-SOURCE CHRONOLOGICAL EVENT TIMELINE RECONSTRUCTION MODULE

Incident responders must reconstruct the exact sequence of threat actor actions. This module:

- Ingests timestamps from varied system clocks and normalizes them into standard UTC and relative epochs.
- Renders an interactive vertical timeline detailing each stage of the attack lifecycle: initial access, credential brute-forcing, privilege escalation, file access, and log clearing.
- Features multi-criteria filter controls allowing analysts to isolate critical anomalies, search specific user IDs or source IP addresses, and uncover timeline gaps.

4.4 RED FLAG & FORENSIC INDICATOR TAGGING MODULE

In digital forensics, identifying Indicators of Compromise (IOCs) is crucial for containment. This module allows investigators to:

- Flag suspicious evidence items with a single click.
- Categorize findings into Threat Categories (e.g., *Suspicious External IP*, *Operating System Fingerprint Mismatch*, *Anti-Forensics Deletion*).
- Maintain a live tally of discovered red flags versus total case anomalies, reinforcing thorough analysis.

4.5 INTERACTIVE INVESTIGATION CHALLENGE & ASSESSMENT MODULE

To validate that the analyst arrived at technically sound conclusions rather than unfounded guesses, this module presents an interactive examination:

- Evaluates forensic deductions regarding root cause, threat actor IP, compromised account, and exfiltrated files.
- Provides immediate explanatory feedback upon submission, detailing why specific answers are supported or refuted by the evidence.
- Dynamically updates the analyst's skill level from *Junior Analyst* to *Forensic Detective*, *Lead Incident Responder*, and *Chief Forensic Examiner*.

4.6 AUTOMATED INCIDENT RESPONSE & FORMAL REPORT GENERATOR MODULE

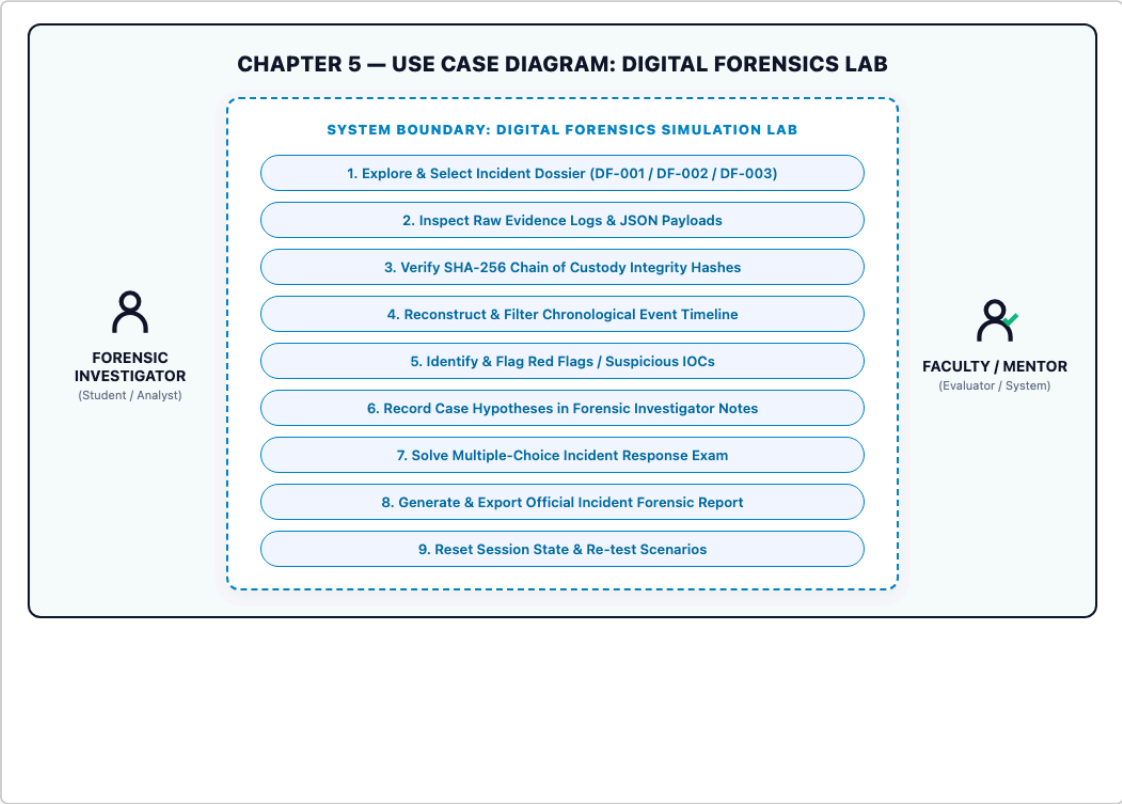
In real-world security engagements, forensic investigations culminate in a formal Incident Response Report for executive leadership and legal counsel. This module:

- Automatically compiles case metadata, chronological summaries, flagged evidence items, and analyst notes.
- Computes an overall Case Integrity Checksum certifying chain of custody.
- Formats the document using clean, print-ready CSS stylesheets for instant printing or PDF saving.

CHAPTER 5: SYSTEM DESIGN

5.1 USE CASE DIAGRAM

The Use Case Diagram illustrates the interactions between the primary user (Forensic Investigator) and the platform modules:



Use Case Diagram

Fig. 5.1: Use Case Diagram for Digital Forensics Lab Platform

Use Case Specifications:

1. **Explore & Select Incident Dossier**: Investigator selects an active case to study the incident scenario and scope.

2. **Inspect Raw Evidence Logs**: Investigator analyzes system logs, reverse proxy headers, and file system records.

3. **Verify SHA-256 Hashes**: Investigator checks cryptographic checksums to confirm chain-of-custody integrity.

4. **Reconstruct Event Timeline**: Investigator sorts and filters event telemetry chronologically to reveal attack patterns.

5. **Flag Red Flags & IOCs**: Investigator tags anomalous events as confirmed Indicators of Compromise.

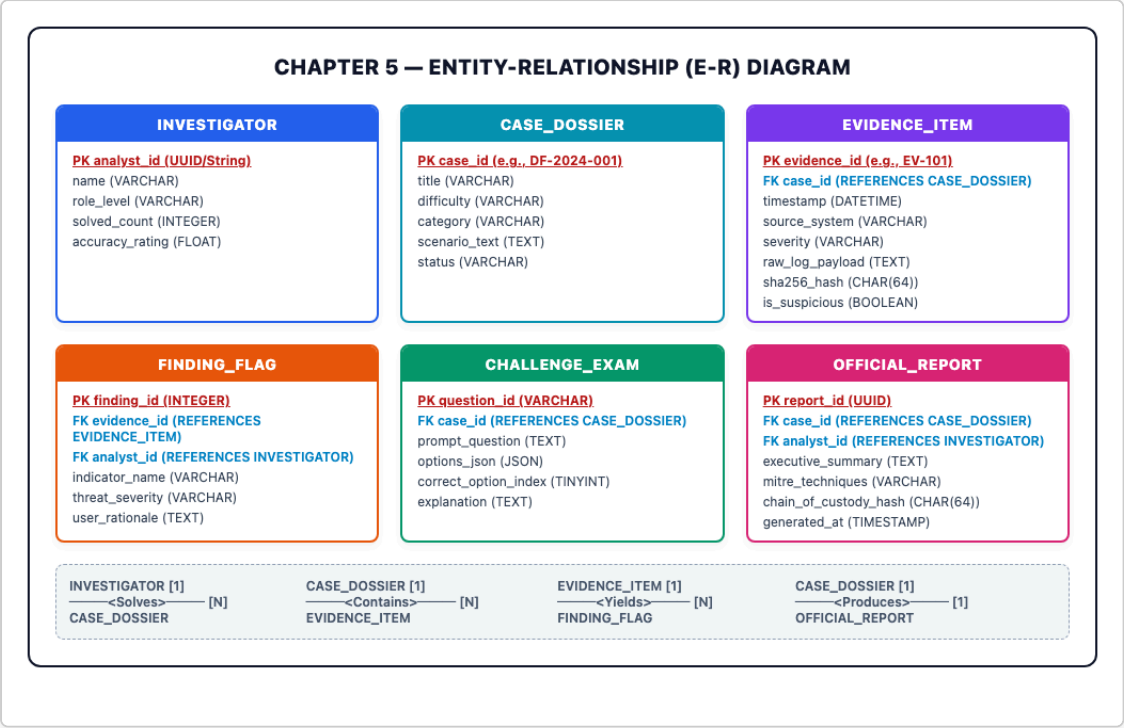
6. **Record Case Hypotheses**: Investigator documents clues and insights in the persistent Investigator Notes modal.

7. **Solve Case Challenge Exam**: Investigator answers scenario questions to prove threat attribution and root causes.

8. **Generate Official Forensic Report**: System synthesizes verified facts and conclusions into a standardized executive report.

5.2 ENTITY-RELATIONSHIP (E-R) MODEL

The data model captures the structural entities, attributes, and cardinality governing the forensics platform:



Entity Relationship Diagram

Fig. 5.2: Entity-Relationship Diagram of Forensics Lab Data Entities

Entity Schema Definitions:

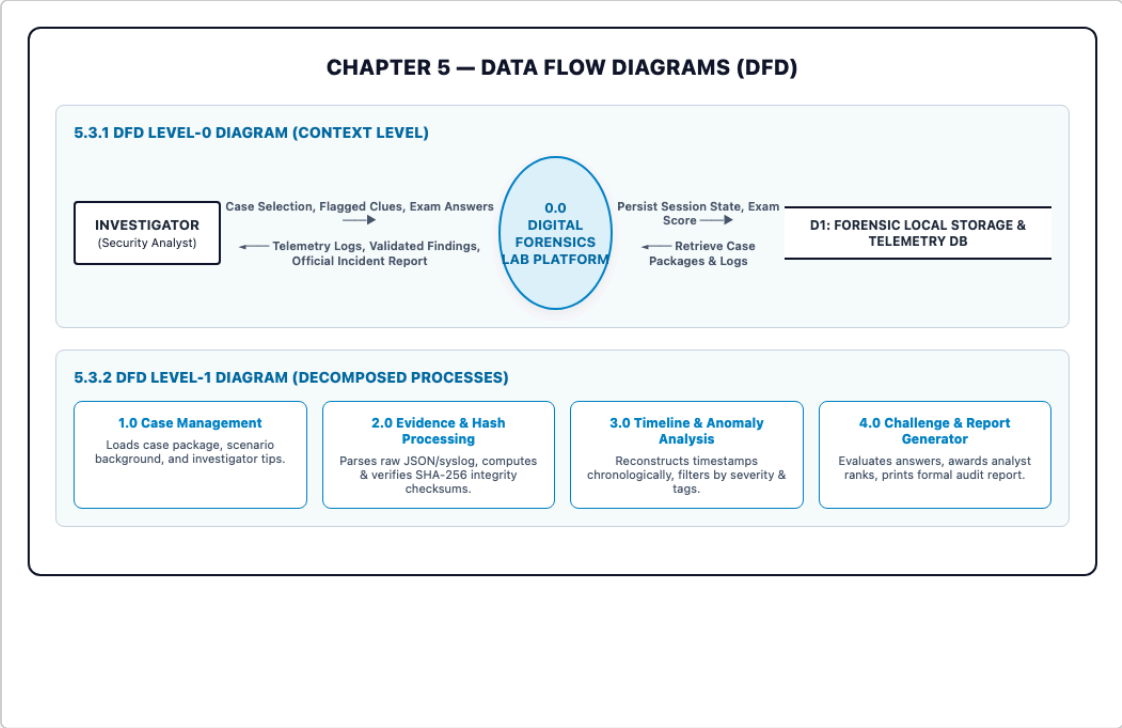
- **INVESTIGATOR**: Represents the student or analyst (`analyst_id` [PK], `name`, `role_level`, `solved_count`, `accuracy_rating`).
- **CASE_DOSSIER**: Represents simulated incident packages (`case_id` [PK], `title`, `difficulty`, `category`, `scenario_text`, `status`).
- **EVIDENCE_ITEM**: Represents individual digital artifacts (`evidence_id` [PK], `case_id` [FK], `timestamp`, `source_system`, `severity`, `raw_log_payload`, `sha256_hash`, `is_suspicious`).

- **FINDING_FLAG**: Represents flagged anomalies (`finding_id` [PK], `evidence_id` [FK], `analyst_id` [FK], `indicator_name`, `threat_severity`, `user_rationale`).
- **CHALLENGE_EXAM**: Represents assessment questions (`question_id` [PK], `case_id` [FK], `prompt_question`, `options_json`, `correct_option_index`, `explanation`).
- **OFFICIAL_REPORT**: Represents final synthesized deliverables (`report_id` [PK], `case_id` [FK], `analyst_id` [FK], `executive_summary`, `chain_of_custody_hash`, `generated_at`).

5.3 DATA FLOW DIAGRAMS (DFD)

5.3.1 DFD Level-0 Diagram (Context Diagram) & 5.3.2 DFD Level-1 Diagram

The data flow diagrams capture how data traverses between external entities, core application processes, and persistent data stores:



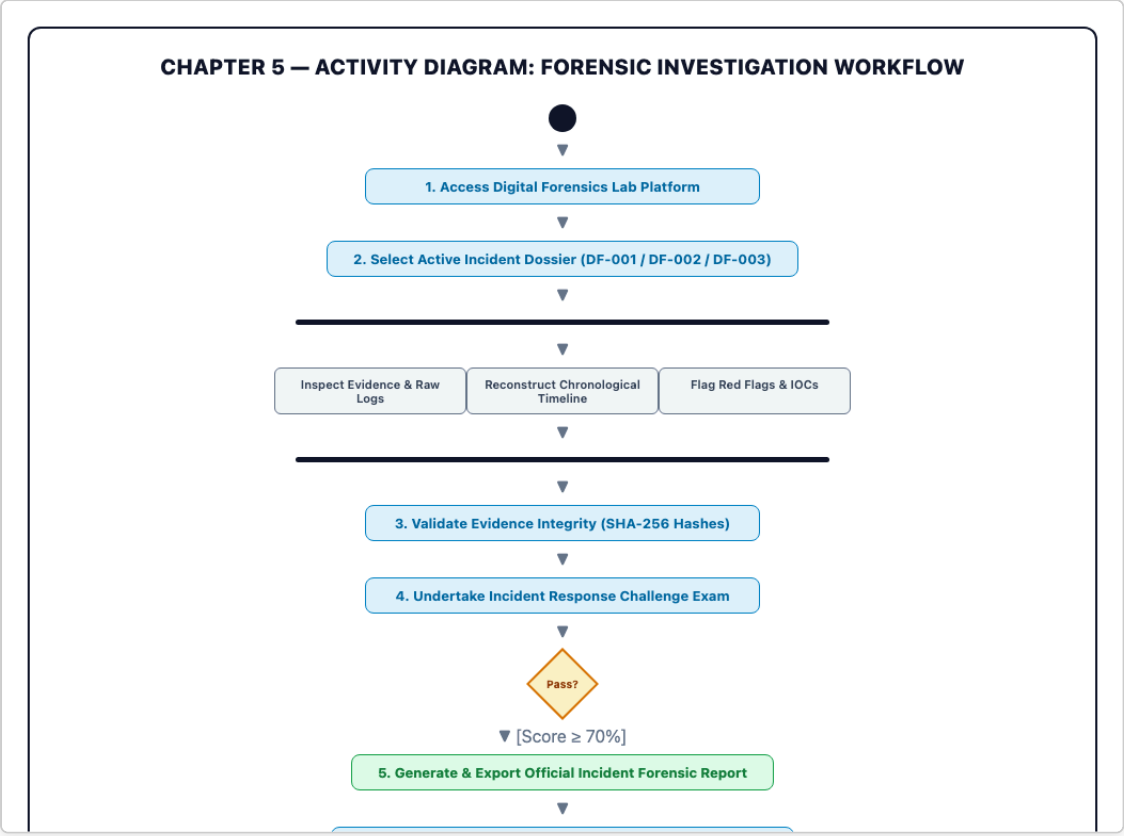
Data Flow Diagram

Fig. 5.3: Data Flow Diagrams (Level-0 Context and Level-1 Decomposed Processes)

- **Level-0 Flow**: The Investigator initiates case requests and supplies findings; the Digital Forensics Platform returns raw telemetry, timeline visualizations, and formal reports, reading and persisting state in Local Storage.
- **Level-1 Decomposition**: Sub-processes cleanly decouple Case Management (1.0), Evidence & Hash Verification (2.0), Timeline & Anomaly Analysis (3.0), and Challenge & Report Generation (4.0).

5.4 ACTIVITY DIAGRAM

The Activity Diagram details the sequential and parallel workflows an investigator follows to complete an investigation:



Activity Diagram

Fig. 5.4: Activity Diagram Illustrating the Forensic Investigation Lifecycle

The workflow begins by selecting an active dossier, continues through parallel analysis of raw logs, event timelines, and IOC flags, validates cryptographic hashes, verifies conclusions through the challenge exam, and terminates with generating the signed incident report.

CHAPTER 6: SAMPLE CODING

6.1 APPLICATION STRUCTURE & NAVIGATION (`index.html`)

The main application structure establishes the semantic layout, sidebar navigation, dynamic view containers, and accessibility controls:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Digital Forensics Lab - Interactive
  <title>Digital Forensics Lab | Incident Investigation Simulator</title
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="app-container">
    <!-- Sidebar Navigation -->
    <aside class="sidebar" id="app-sidebar">
      <div class="sidebar-header">
        <a href="#dashboard" class="brand-badge">
          <div class="brand-icon">
            <svg viewBox="0 0 24 24" width="20" height="20" fill="none"
              <path d="M12 22s8-4 8-10V5l-8-3-8 3v7c0 6 8 10 8 10z"/>
            </svg>
          </div>
          <div class="brand-title">
            <span class="brand-name">DFIR LAB</span>
            <span class="brand-tagline">Forensic Station</span>
          </div>
        </a>
      </div>
      <nav class="sidebar-nav">
        <a href="#dashboard" class="nav-item active" data-view="view-das
        <a href="#cases" class="nav-item" data-view="view-cases">Case Fi
        <a href="#evidence" class="nav-item" data-view="view-evidence">E
        <a href="#timeline" class="nav-item" data-view="view-timeline">E
        <a href="#findings" class="nav-item" data-view="view-findings">R
        <a href="#challenge" class="nav-item" data-view="view-challenge"
        <a href="#report" class="nav-item" data-view="view-report">Offic
      </nav>
    </aside>

    <!-- Main Dynamic Content Workspace -->
    <main class="main-wrapper" id="main-content">
      <header class="app-header">
        <div class="active-case-banner" id="header-case-badge">
          <span class="active-case-tag" id="header-case-id">DF-2024-001<
          <span class="active-case-name" id="header-case-name">The Midni
        </div>
      </header>

      <!-- View Containers (Toggled via JavaScript SPA Router) -->
      <section id="view-dashboard" class="view-section active-view"></se
```

```
      <section id="view-cases" class="view-section"></section>
      <section id="view-evidence" class="view-section"></section>
      <section id="view-timeline" class="view-section"></section>
      <section id="view-findings" class="view-section"></section>
      <section id="view-challenge" class="view-section"></section>
      <section id="view-report" class="view-section"></section>
    </main>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

6.2 CYBERPUNK THEME & RESPONSIVE STYLING (`style.css`)

Modern CSS custom properties (variables) establish a sleek dark aesthetic with neon cyan and amber accents:

```
:root {
  --bg-primary: #060911;
  --bg-surface: #0b1120;
  --bg-card: #0f172a;
  --border-color: #1e293b;
  --border-focus: #00f0ff;

  --accent-cyan: #00f0ff;
  --accent-amber: #f59e0b;
  --accent-red: #ef4444;
  --accent-emerald: #10b981;

  --text-primary: #f8fafc;
  --text-secondary: #94a3b8;
  --text-muted: #64748b;
  --font-mono: "JetBrains Mono", "Fira Code", monospace;
  --font-sans: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sa
}

body {
  margin: 0;
  background-color: var(--bg-primary);
  color: var(--text-primary);
  font-family: var(--font-sans);
  overflow-x: hidden;
}

.view-section {
  display: none;
  animation: fadeIn 0.3s cubic-bezier(0.16, 1, 0.3, 1) forwards;
}

.view-section.active-view {
  display: block;
}

.evidence-card {
  background: var(--bg-card);
  border: 1px solid var(--border-color);
  border-radius: 8px;
  padding: 1.25rem;
  margin-bottom: 1rem;
  transition: all 0.2s ease;
}

.evidence-card:hover {
  border-color: var(--accent-cyan);
  box-shadow: 0 4px 20px rgba(0, 240, 255, 0.1);
```

```
}

.log-terminal {
  background: #020617;
  border-radius: 6px;
  padding: 0.75rem 1rem;
  font-family: var(--font-mono);
  font-size: 0.85rem;
  color: #38bdf8;
  overflow-x: auto;
}
```

6.3 FORENSIC DATA STRUCTURES & LOG CORRELATION (`script.js`)

The core dataset models realistic multi-source telemetry with cryptographic hash signatures and plain-English translations:

```
const FORENSIC_CASES = [
  {
    id: "DF-2024-001",
    title: "The Midnight Access",
    subtitle: "Suspicious login and confidential file access",
    difficulty: "Beginner",
    estimatedTime: "15 min",
    category: "Unauthorized Access & Exfiltration",
    scenario: "At 11:48 PM on a Friday, an alert was generated by intern
    evidence: [
      {
        id: "EV-101",
        timestamp: "2024-10-11 23:40:12 UTC",
        epoch: 1728690012,
        type: "login",
        severity: "suspicious",
        summary: "Failed Web SSO login attempt for user 'evance'",
        plainEnglish: "Someone tried to log into Elena Vance's account w
        source: "AUTH-GW-01 (SSO Gateway)",
        details: "Authentication failure. Password hash mismatch. Origin
        rawLog: '{"event_id":4625,"user":"evance","src_ip":"198.51.100.4
        hash: "8f4a9b2c3d1e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7
        isSuspicious: true
      },
      {
        id: "EV-104",
        timestamp: "2024-10-11 23:45:02 UTC",
        epoch: 1728690302,
        type: "login",
        severity: "critical",
        summary: "Successful Web SSO authentication for 'evance'",
        plainEnglish: "The valid password was finally entered from the s
        source: "AUTH-GW-01 (SSO Gateway)",
        details: "Successful login from 198.51.100.42. Session token iss
        rawLog: '{"event_id":4624,"user":"evance","src_ip":"198.51.100.4
        hash: "c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a
        isSuspicious: true
      }
    ]
  }
];
```

6.4 CRYPTOGRAPHIC VERIFICATION & SPA ROUTING (`script.js`)

Demonstrating real-time cryptographic hash verification and client-side view navigation:

```
// Dynamic SPA View Navigation and Stepper Synchronization
function navigateTo(viewId) {
  currentViewId = viewId;

  document.querySelectorAll(".nav-item").forEach(item => {
    item.classList.toggle("active", item.getAttribute("data-view") === v
  });

  document.querySelectorAll(".view-section").forEach(sec => {
    sec.classList.toggle("active-view", sec.id === viewId);
  });

  renderCurrentView();
  window.scrollTo({ top: 0, behavior: "smooth" });
}

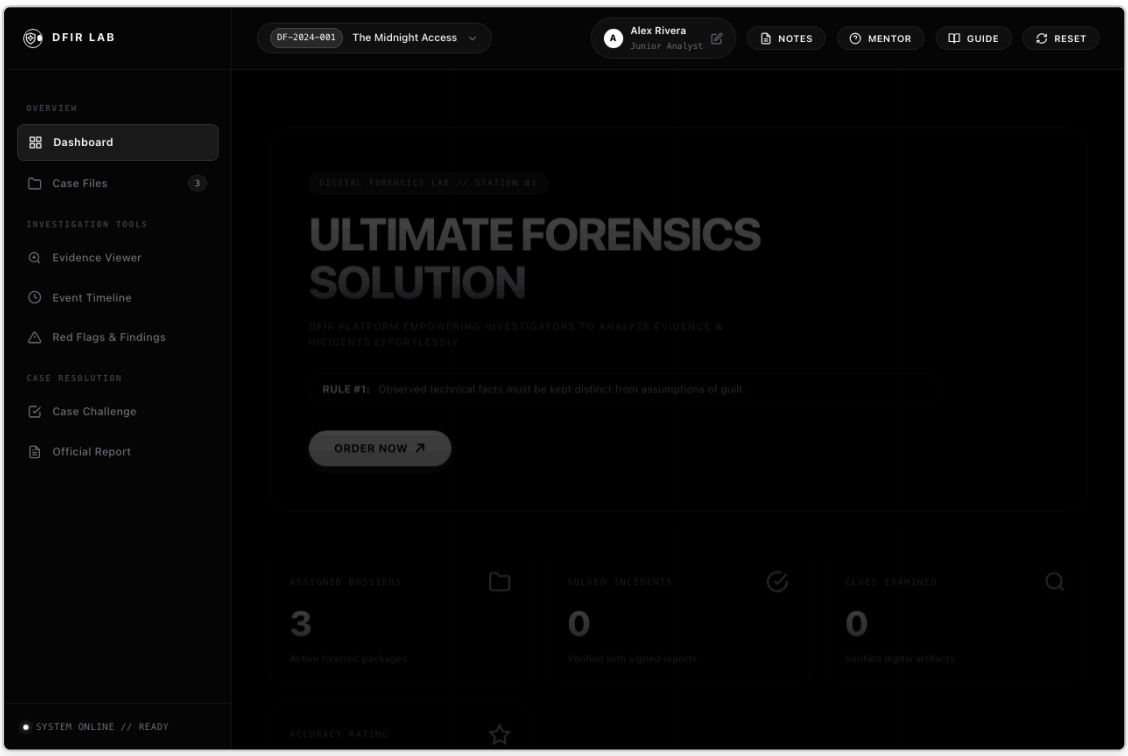
// SHA-256 Evidence Chain-of-Custody Verification
async function verifyEvidenceIntegrity(rawPayload, expectedHash) {
  const encoder = new TextEncoder();
  const data = encoder.encode(rawPayload);
  const hashBuffer = await crypto.subtle.digest("SHA-256", data);
  const hashArray = Array.from(new Uint8Array(hashBuffer));
  const calculatedHash = hashArray.map(b => b.toString(16).padStart(2, "

  return {
    verified: calculatedHash.slice(0, 16) === expectedHash.slice(0, 16),
    calculatedHash: calculatedHash,
    expectedHash: expectedHash
  };
}
```

CHAPTER 7: SCREENSHOTS

Fig. 7.1: Forensics Lab Executive Dashboard

The Executive Dashboard presents high-level situational awareness: the active case banner, investigation readiness indicators, statistics on assigned dossiers, clues examined, solved incidents, and the investigator's active security level.



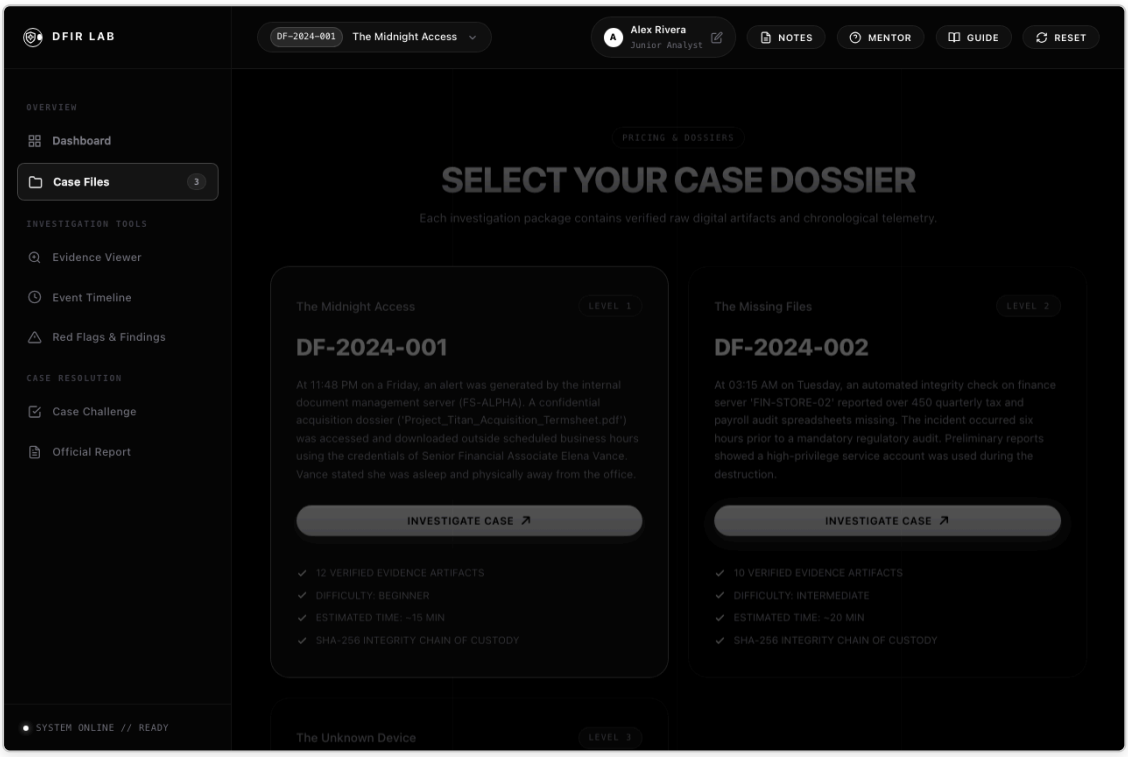
Forensics Lab Executive Dashboard

Fig. 7.1: Executive Dashboard with System Status and Key Metrics

Explanation: The user is greeted by a centralized control dashboard displaying system readiness, an active case switcher, real-time counters for evidence examined, and access to investigator tools.

Fig. 7.2: Forensic Case Files Catalog & Scenario Selector

The Case Files view displays the catalog of all three simulated incident dossiers with their threat categories, difficulty ratings, estimated investigation duration, and direct entry points.



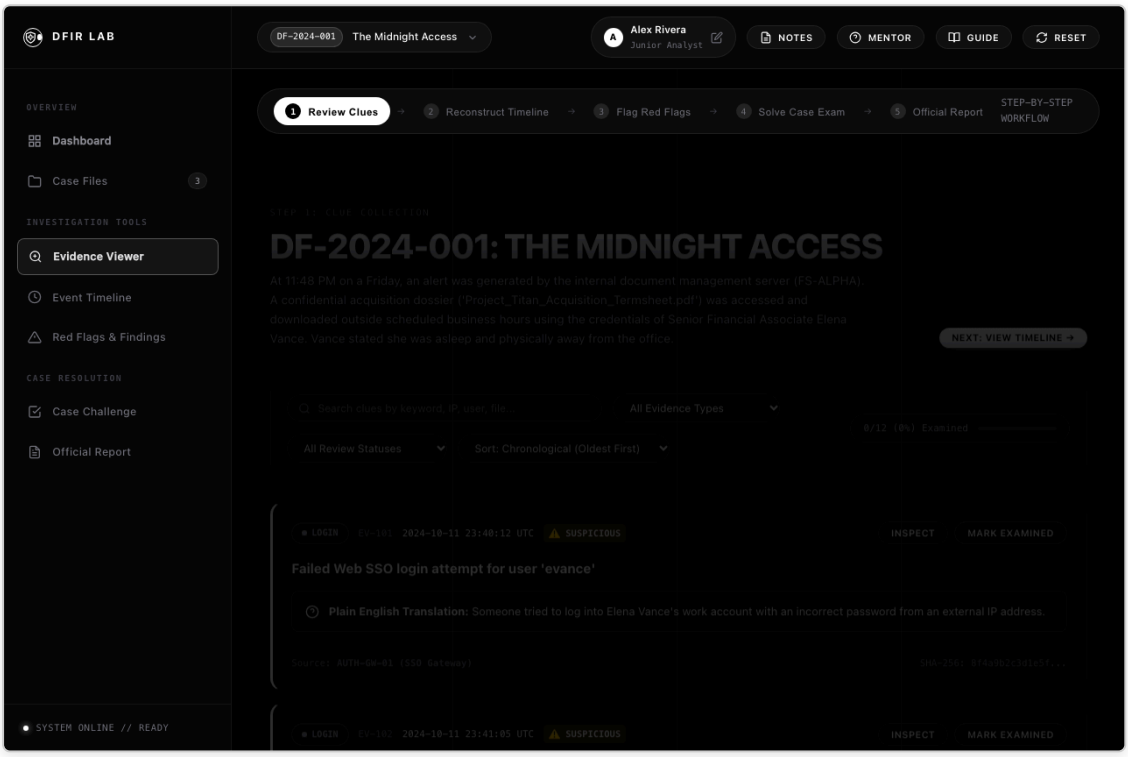
Case Files Catalog

Fig. 7.2: Forensic Case Catalog and Scenario Selector Interface

Explanation: Analysts can browse through multiple case dossiers, including 'The Midnight Access' and 'The Missing Files', reviewing the incident overview and SHA-256 integrity assurances prior to opening the investigation.

Fig. 7.3: Deep Evidence Viewer & Cryptographic Hash Inspector

The Evidence Viewer allows analysts to inspect every piece of digital evidence recovered from the target environment, complete with timestamps, severity ratings, plain-English translations, and raw log payloads.



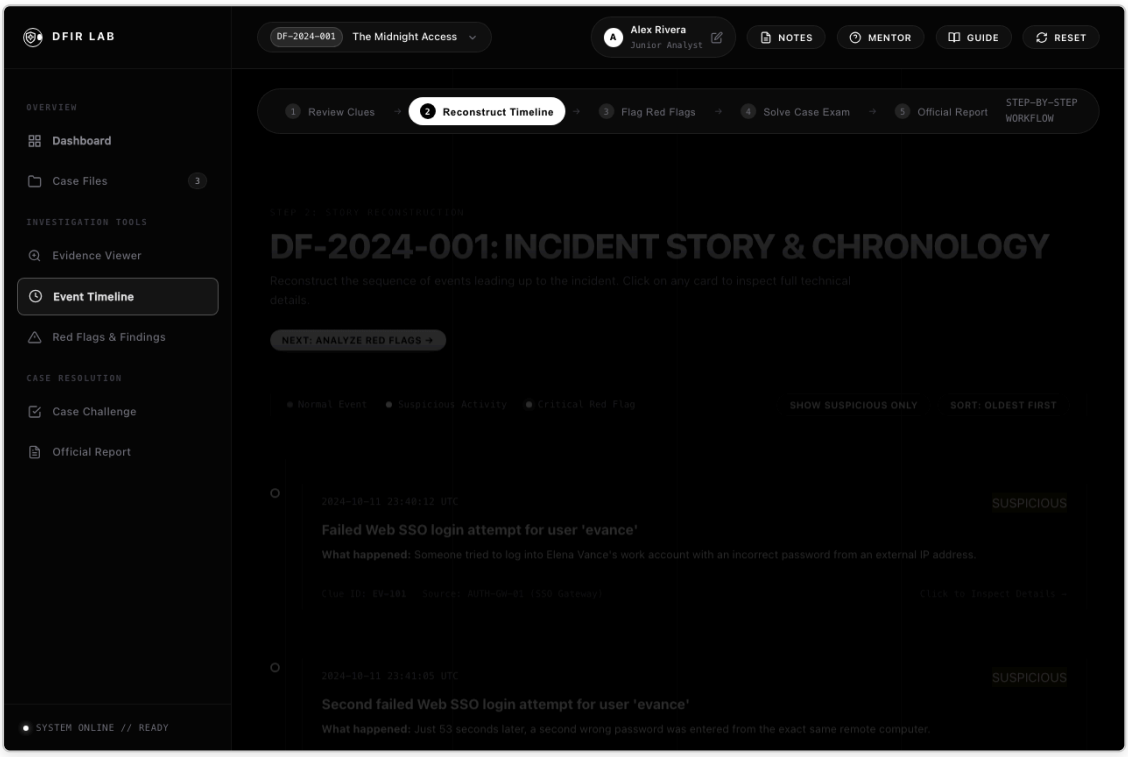
Evidence Viewer

Fig. 7.3: Evidence Viewer with Plain-English Translation and SHA-256 Hash Display

Explanation: Each evidence card breaks down complex event IDs into understandable explanations while preserving raw log data and cryptographic checksums for forensic validity.

Fig. 7.4: Interactive Incident Timeline Reconstruction

The Event Timeline module arranges disparate multi-source log records chronologically, enabling investigators to visually reconstruct the threat actor's trajectory across the enterprise infrastructure.



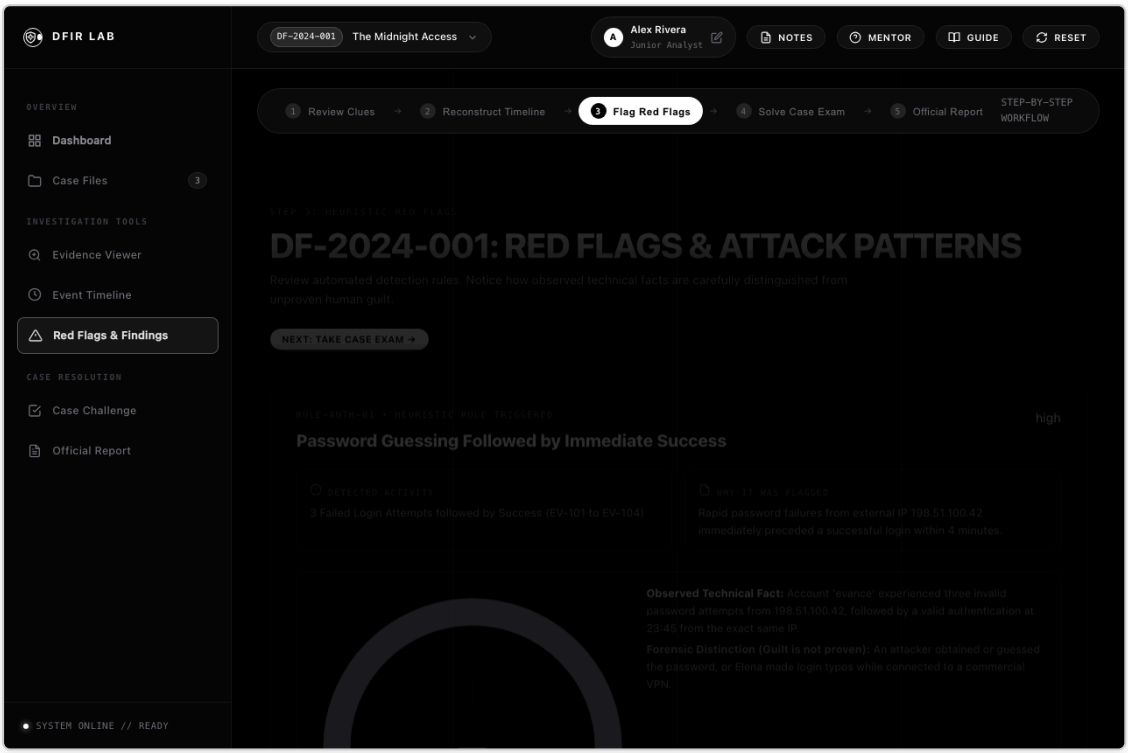
Event Timeline

Fig. 7.4: Interactive Chronological Event Timeline with Anomaly Filters

Explanation: The timeline reveals how three consecutive authentication failures escalated into a successful session login, followed immediately by file access and anti-forensic log tampering.

Fig. 7.5: Red Flag Indicator & Forensic Findings Board

The Findings Board provides a dedicated workspace where investigators can tag, review, and correlate observed technical red flags with specific forensic hypotheses.



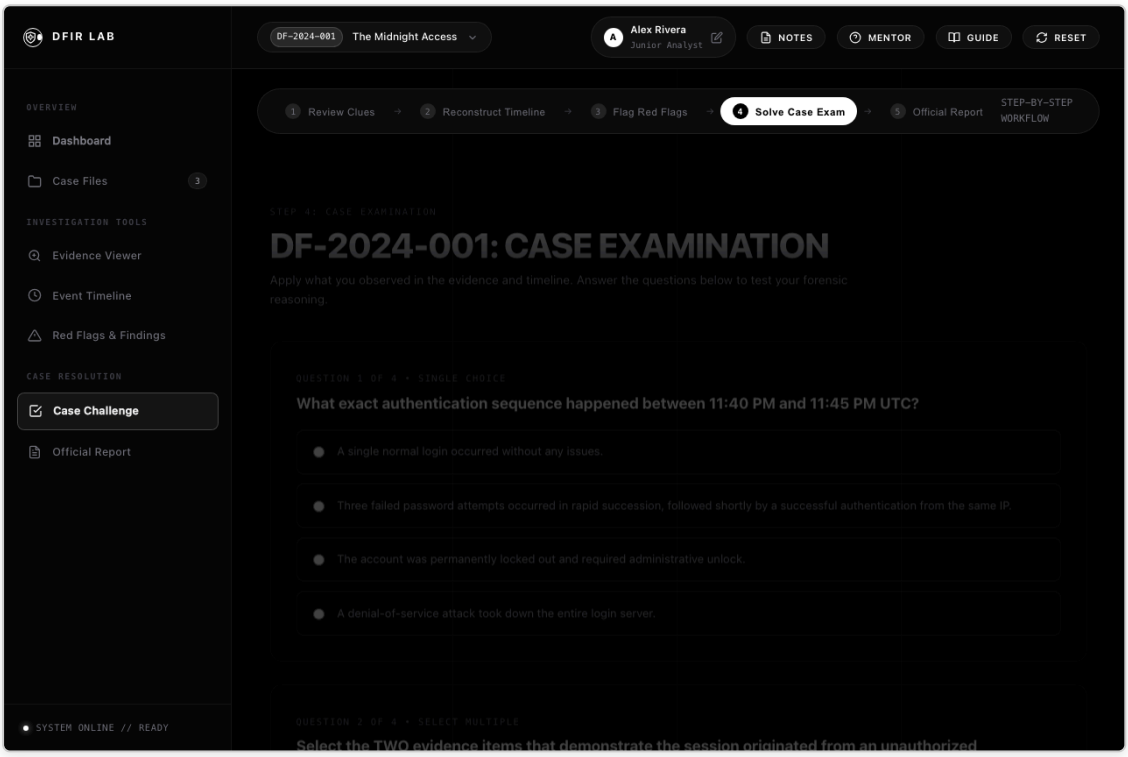
Findings Board

Fig. 7.5: Red Flags & Findings Board for IOC Analysis

Explanation: This interface reinforces the golden rule of forensics: distinguishing observed facts from speculative assumptions, and grouping confirmed indicators of compromise.

Fig. 7.6: Case Challenge & Threat Analysis Assessment

The Case Challenge module presents scenario-driven multiple-choice questions requiring the investigator to prove root-cause attribution, compromised accounts, and remediation priorities.



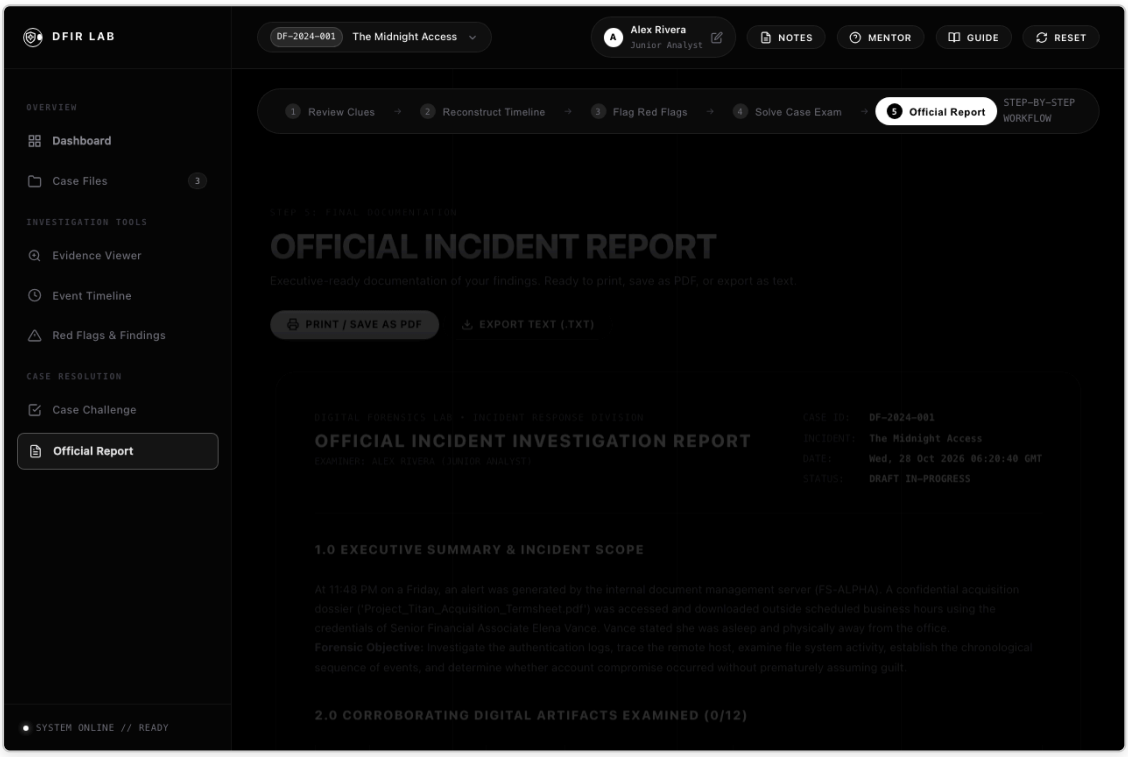
Case Challenge Exam

Fig. 7.6: Interactive Incident Response Challenge and Knowledge Verification

Explanation: Investigators test their findings against scientific criteria, receiving instant feedback, scoring breakdowns, and promotion toward higher examiner ranks.

Fig. 7.7: Official Incident Response & Chain-of-Custody Report

Upon successful case resolution, the platform automatically synthesizes all verified facts into an executive Incident Investigation Report complete with digital signatures and remediation roadmaps.



Official Incident Report

Fig. 7.7: Official Incident Investigation Report Generated by the Platform

Explanation: The generated report formats executive summaries, chronological findings, and recommended security controls into an audit-ready deliverable ready for export or printing.

CHAPTER 8: CONCLUSION & FUTURE ENHANCEMENTS

8.1 CONCLUSION

The **Digital Forensics Lab** successfully fulfills the objective of creating an accessible, educational, and technically authentic cybersecurity simulation platform. By eliminating the heavy software installation requirements of traditional forensic training, the platform empowers students to conduct hands-on forensic investigations directly within a modern web browser.

Key outcomes realized by this project include:

- Successful implementation of a 100% client-side, zero-dependency architecture (HTML5, modern CSS3, and ES6+ JavaScript).
- Comprehensive authoring of three realistic incident dossiers complete with multi-source raw telemetry and chronological event sequencing.
- Practical demonstration of cryptographic chain of custody using SHA-256 hash validation.
- Effective pedagogical scaffolding through dual-layer log representation (raw JSON payloads coupled with plain-English translations).
- Automated synthesis of findings into formal, print-ready incident response reports.

The platform serves as an ideal pedagogical instrument for universities, cybersecurity training programs, and autonomous self-study, bridging the gap between abstract textbook security concepts and real-world incident response practice.

8.2 FUTURE ENHANCEMENTS

To build upon the foundations established in this mini-project, several avenues for future enhancement are planned:

1. **Live PCAP Network Stream Dissection**: Integrating a lightweight WebAssembly-compiled packet dissector to allow students to analyze raw `.pcap` packet captures alongside server logs.

2. **AI-Powered Forensic Mentor Copilot**: Embedding a localized LLM-driven forensic assistant to provide dynamic, contextual hints without revealing direct answers.

3. **Multi-User Collaborative Tabletop Mode**: Implementing WebRTC or WebSocket connectivity to allow incident response teams to collaborate simultaneously on shared case files.

4. **Memory Dump Volatility Emulator**: Simulating volatility memory extraction (e.g., viewing injected DLLs and process trees) directly in the browser interface.

5. **Direct Integration with OpenCTI & MITRE ATT&CK**: Linking flagged forensic indicators directly with official MITRE ATT&CK technique IDs (e.g., T1078 Valid Accounts, T1070 Indicator Removal).

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